ME 5309 – Mid-Semester Examination

Answer Booklet

Instructions: Attempt any three questions. Each question carries 10 marks. Show working where appropriate.

Question 1 [10 marks]

Q1(a) [4 marks]

The pixel pitch is 0.005 mm, so f_x = 16/0.005 = 3200 pixels. Sensor width = 6.4 mm. Belt field of view = 400x6.4/16 = 160 mm; spatial resolusion = 160/1280 = 0.125 mm/pixel.

Q1(b) [3 marks]

ADC step = 5/4096 = 1.2207 mV. I divide by 0.1 mV/N, giving ΔF = 12.207 N.

Q1(c) [3 marks]

Question 3 [10 marks]

Q3(a) [2 marks]

The estimate is noisy and can fluctuate from sample to sample.

Q3(b) [2 marks]

For H(s)=ωc/(s+ωc), ωc=2π(20)=125.664 rad/s and T=0.001 s. Tustin gives H(z)=α(1+z^{-1})/[(1+α)+(α-1)z^{-1}], α=ωcT/2=0.0628319. Thus:

$$y[k] = 0.881765y[k-1] + 0.0591175(x_k + x[k-1])$$

DC gain = (2x0.0591175)/(1-0.881765) approximately 1.000.

$\omega_c = 2\pi(20) = 125.664$ rad/s. With $T = 0.001$ s, $\alpha = \omega_c T/2 = 125.664(0.001)/2 = 0.062832$. The recurrence coefficients are approximately 0.881765 on the previous output and 0.0591175 on each input sample. Check: $2(0.0591175)/(1 - 0.881765) \approx 1$, so the DC gain is close to one.

Q3(c) [4 marks]

A-LC=[[-ℓ_1,1], [-ℓ_2,-2]]. I obtain p(s)=s^2+(ℓ_1+2)s+(ℓ_1+ℓ_2). Matching $(s+10)^2$ gives ℓ_1=18 and ℓ_2=82.

Q3(d) [2 marks]

The observer is simpler because it only smooths the measured velocity. The filter is safer when the model is poor.

Question 4 [10 marks]

Q4(a) [2 marks]

The characteristic polynomial is s^2+4s+8, so I report poles -4 and -8. Unity DC gain requires Nff=8.

Q4(b) [2 marks]

Q4(c) [4 marks]

The desired polynomial is $(s+4)(s^2+4s+8)$. The closed-loop polynomial should be cubic because an integrator is added.

Q4(d) [2 marks]

Integral action helps eliminate error. A problem is that the controller can become unstable.